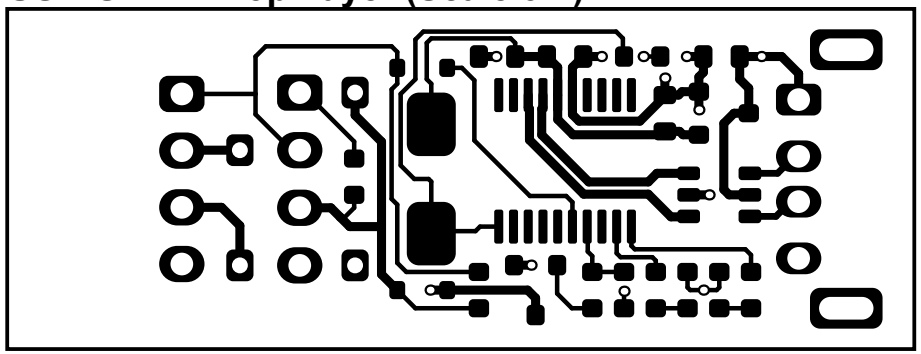
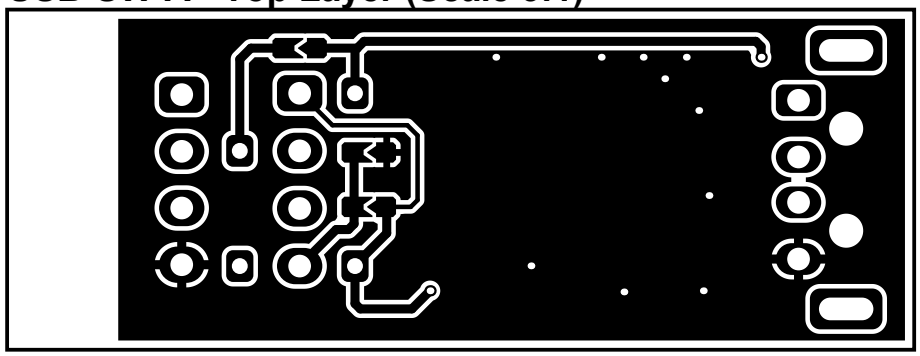
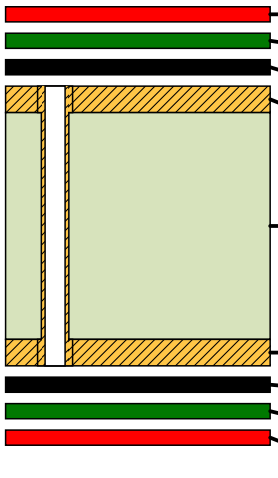
USB-SW-A - Top Layer (Scale 3:1)



USB-SW-A - Top Layer (Scale 3:1)



Layer Stack Legend



Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.0100mm	SM-001	Solder Mask	GTS
PbSn	Top Copper Plating	0.0200mm		Surface Finish	
CF-004	**Top Layer**	**0.0350mm**		**Signal**	**GTL**
Core		***1.5000mm***	***FR-4 TG140***	***Dielectric***	
CF-004	**Bottom Layer**	**0.0350mm**		**Signal**	**GBL**
PbSn	Bottom Copper Plating	0.0200mm		Surface Finish	
Surface Material	Bottom Solder	0.0100mm	SM-001	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

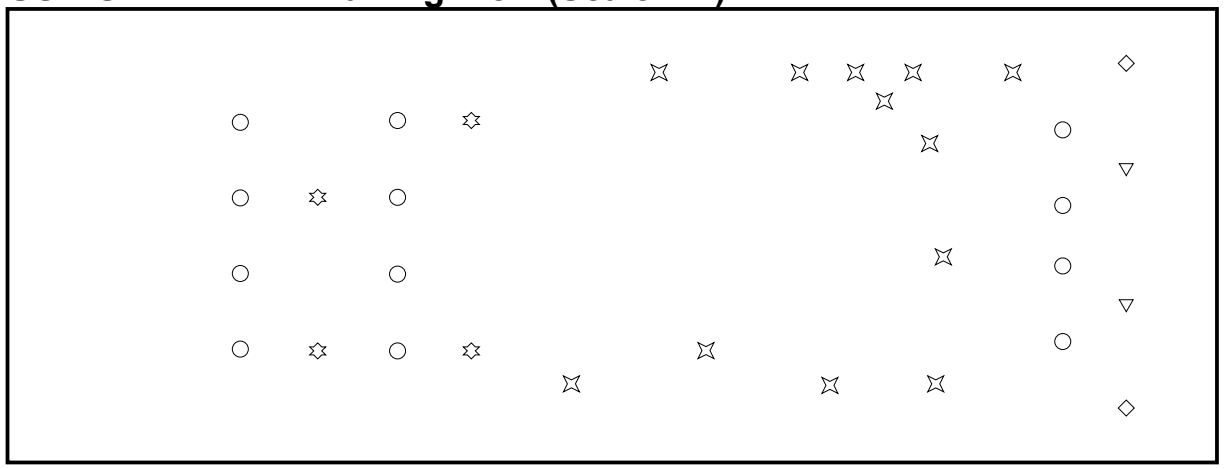
Total thickness: 1.6300mm

Board Specification

Minimum Track	0.2 mm	Solder Mask Color	**Blue**
Minimum Gap	0.2 mm	Silkscreen	**White**
VIA Process	Tented	Board Finish	HASL - LEAD

Project: *USB controlled Relay*		Revision: *A*
Title: **Board Specifications**		
Sheet: ~~=sheetname~~ of ~~=sheetcount~~	Author: Alfredo Cortellini	
Date: Oct-01-2024	License: CC BY-NC-SA 4.0	

USB-SW-A - Drill Drawing View (Scale 4:1)

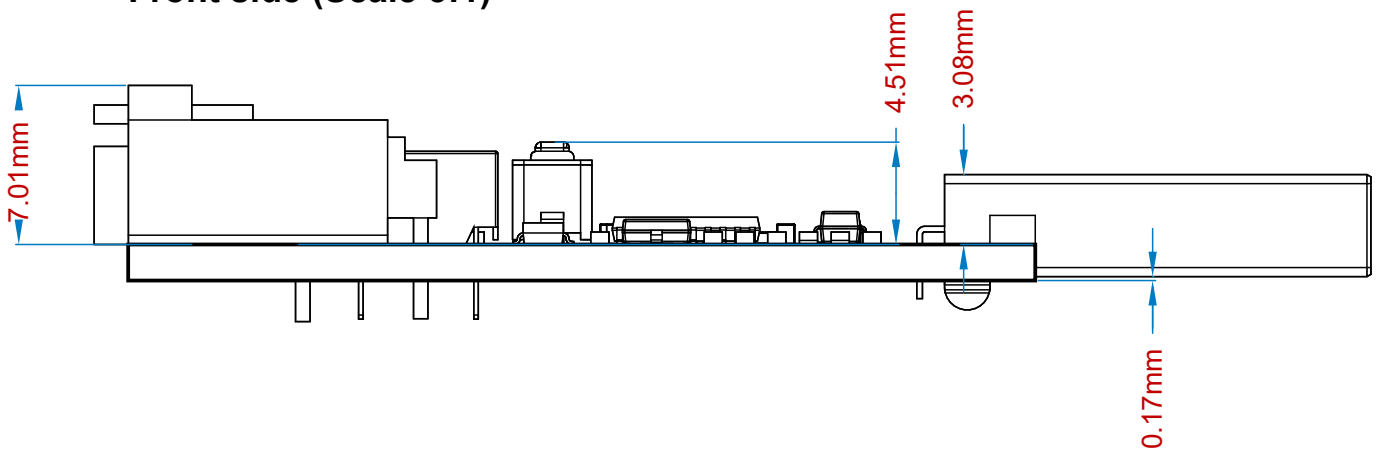


Drill Table

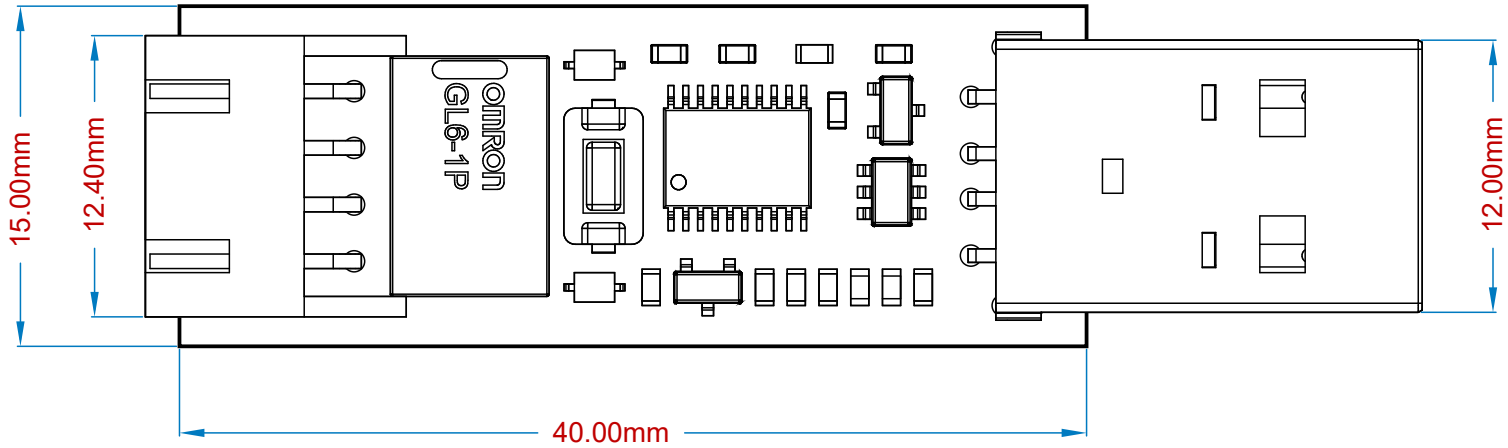
Symbol	Count	Hole Size	Plated	Hole Type	Via / Pad	Hole Tolerance
☒	12	0.30mm	Plated	Round	Via	
✡	4	0.70mm	Plated	Round	Pad	
○	12	1.00mm	Plated	Round	Pad	
▽	2	1.10mm	Non-Plated	Round	Pad	
◇	2	2.40mm	Plated	Slot	Pad	
	32 Total					

Project: *USB controlled Relay*		Revision: *A*
Title: **Board Specifications**		
Sheet: 1 of 1	Author: Alfredo Cortellini	
Date: Oct-01-2024	License: CC BY-NC-SA 4.0	

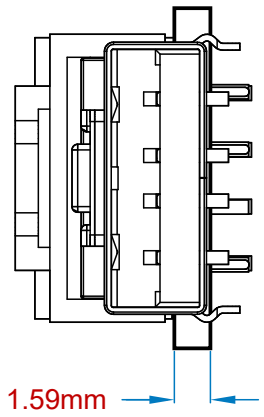
Front side (Scale 3:1)



USB-SW-A - View from Top side (Scale 3:1)

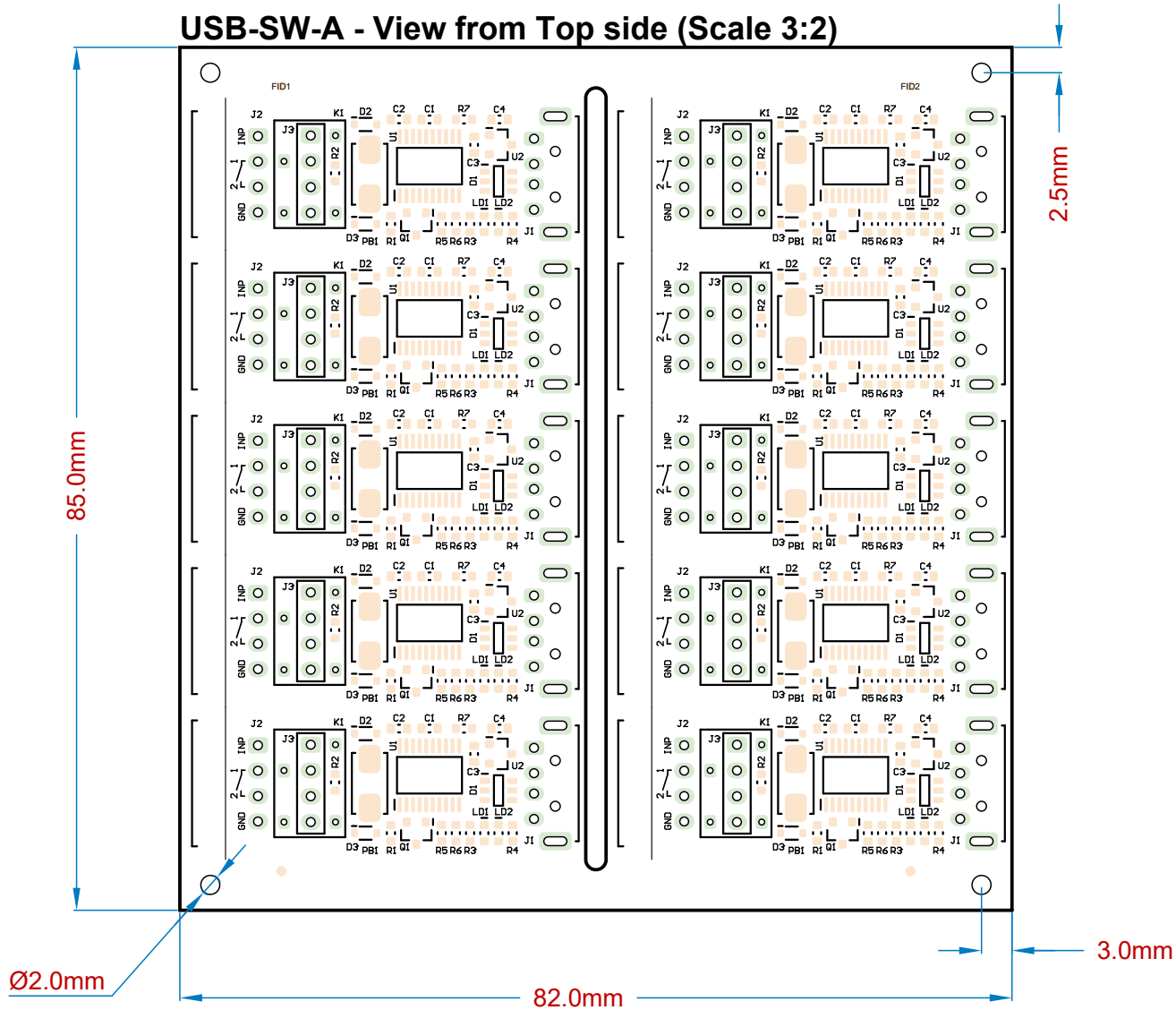


Right side (Scale 3:1)



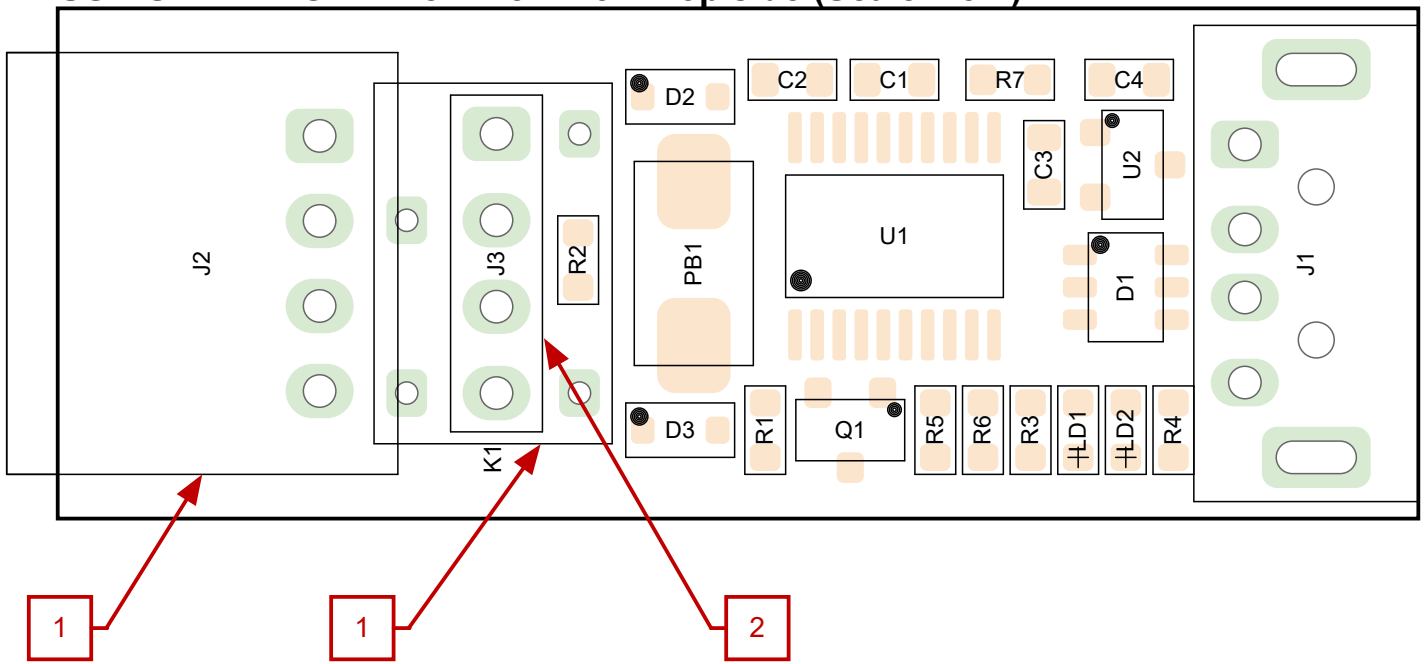
Project: *USB controlled Relay*		Revision: *A*
Title: **Board Dimensions**		
Sheet: ~~=sheet~~ of ~~=sheets~~	Author: Alfredo Cortellini	
Date: Oct-01-2024	License: CC BY-NC-SA 4.0	

USB-SW-A - View from Top side (Scale 3:2)



Project: *USB controlled Relay*	Revision: *A*
Title: **Panel Dimensions**	
Sheet: ~~=shortname~~ of ~~=sheetcount~~	Author: Alfredo Cortellini
Date: Oct-01-2024	License: CC BY-NC-SA 4.0

USB-SW-A - BOM Line - View from Top side (Scale 4.5:1)



Bill Of Materials

Line #	Designator	Quantity	Manufacturer Part Number
	PB1	1	TL3302AF180QJ
	Q1	1	BSS138K-13
	LD1	1	LTST-C191KRKT
	LD2	1	LTST-C191KGKT
	U1	1	STM32F042F4P6
	U2	1	AP7313-33SRG-7
	R1, R2, R6, R7	4	RC0603FR-0710KL
	R3, R4, R5	3	RC0603FR-07470RL
	D1	1	USBLC6-4SC6
	D2, D3	2	MBR0530T1G
	C1	1	CGA3E2X7R1H104K080AA
	C2, C3, C4	3	C1608X7R1C105K080AC
	J3	1	70553-0038
	K1	1	G6L-1P-DC5
	J1	1	48037-0001
	J2	1	S4B-XH-A(LF)(SN)

Assembly Notes

1. Install K1 and J2 if you want to use the relay ON-Off feature. Do NOT install J3
2. Install J3 if you want to use the fan control feautre. Follo the instructions on the back if you have a regular or a PWM fan. Do NOT install K1 and J2

Project: *USB controlled Relay*		Revision: *A*
Title: **Board Assembly**		
Sheet:	Author: Alfredo Cortellini	
Date: Oct-01-2024	License: CC BY-NC-SA 4.0	